Thomas Crowe

Senior Software Engineer

Full-Stack Development • React • Node/Express • PostgreSQL • PHP • Python

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SUMMARY

I'm a full-stack engineer who likes building software that takes real work off people's plates. I work alongside the people who use what I build, find where their day gets stuck, and turn that into tools they rely on. I wear many hats, moving between product, integrations, infrastructure, and AI agents as the work needs, and I use AI coding tools to move faster while designing and reviewing every change. My automation has generated 3,800+ NIST-traceable certificates and eliminates 500+ hours of manual work per year.

EXPERIENCE

Senior Software Engineer | CalibrationWands.com (Fastec Services LLC) Feb 2024 – Present

Travelers Rest, SC

Design, build, and ship production software for a B2B manufacturer of NIST-traceable test standards used by food, pharmaceutical, and textile quality-control teams.

- **Test Standards Hub** — built an internal full-stack platform (React/Vite, Express, Prisma/PostgreSQL) that turns WooCommerce orders into manufacturing outputs: 3,800+ auto-generated NIST-traceable certificates, engraving SVGs, and STL files sent to a CNC machine. Certificates are AES-256 locked and cryptographically signed (pyHanko) for tamper-evident verification, and customers view and download them from their CalibrationWands.com account dashboard via QR codes on packing slips, printed certificates, and product labels.
- **WooCommerce platform engineering** — 50+ custom plugins: Stripe checkout guards, pay-on-PO terms with credit limits, quote-to-order, certificate portal, schema.org SEO; feature-flagged deploys with automatic rollback. Ported the SKU generator to PHP with byte-identical output across all 7,093 product configurations.
- **Custom storefront engineering** — dynamic SKU-configurator across 10 product lines (JavaScript + PHP WooCommerce hooks) generating valid SKUs in real time, with cart, order-meta, email, and PDF integration; debugged a cache-induced add-to-cart race condition; Dompdf invoice templates, Stripe payments, Playwright end-to-end tests.
- **Accounting & inventory integration** — connected the order platform to QuickBooks Desktop (Web Connector/qbXML) for invoices, estimates, items, customers, and inventory adjustments, with per-write verification, an audit change log with revert, and a provider-agnostic accounting interface built for a future system migration; integrated Zoho Inventory so stock is decremented and builds are planned automatically as certificates are generated.
- **Zero-downtime production migration** — migrated a live B2B store to dedicated DigitalOcean NVMe hosting with no downtime or data loss, cutting recurring outages from 145+/month to near zero (Cloudflare DNS/CDN, Redis Object Cache Pro, restored SPF/DKIM/DMARC), then added a Git-deployed Bedrock staging environment for safe pre-production testing.

PROJECTS

- **FlowLocal** — Open-source local voice-dictation stack for Windows — GPU faster-whisper transcription, Ollama text cleanup, retrieval over personal notes, and streamed Kokoro TTS replies (FastAPI), fully local. github.com/wtcrowe4/FlowLocal
- **Agent Workstation** — Daily-driver tooling for directing AI coding agents: DialDeck Surface Dial control surface for herdr agent orchestration (AutoHotkey v2), a local Kokoro TTS daemon (FastAPI), and a herdr message-inbox plugin. github.com/wtcrowe4/DialDeck

SKILLS

Languages: JavaScript/TypeScript, Python, PHP, SQL, Bash, PowerShell, C#, HTML/CSS
Web: React, Next.js, Node.js/Express, REST APIs, Three.js/React Three Fiber, Tailwind, Vite
Data: PostgreSQL, MySQL/MariaDB, SQL Server, MongoDB, Prisma, pgvector, Python ETL, FastAPI
Testing & Quality: Vitest, Node test runner, Supertest, Playwright (E2E), ESLint, Git code review, CI test gates
Infrastructure & DevOps: Docker, Linux/systemd, GitLab CI, GitHub, Cloudflare (DNS/CDN/Tunnel/Access), Google Cloud (GCP), AWS, DigitalOcean, Oracle Cloud, Nginx, Sentry, Tailscale, SOPS/age
Business Systems: WordPress/WooCommerce, Stripe, QuickBooks Desktop (qbXML), Zoho Inventory, UPS API, PrintNode, Google Workspace
AI & Automation: vLLM, Ollama, LangGraph/LangChain, RAG (pgvector), PyTorch, Unsloth fine-tuning, AWQ/GPTQ, Langfuse, Whisper, MCP, Claude Code agents, CodeRabbit, pyHanko

ADDITIONAL EXPERIENCE

General Manager, Arnold's of Powdersville — Powdersville, SC 2016 – 2024

Daily operations, budgeting, inventory, and team leadership for a high-volume restaurant; Clover POS administration and troubleshooting.

EDUCATION

North Carolina State University — Software Development Program, Computer Software Engineering 2022
Clemson University — Computer Engineering 2010 – 2013
College of Charleston — Honors College, Biochemistry 2007 – 2010